16.1 Fraction Basics - Worksheet 1

1 Convert $\frac{15}{4}$ from an improper fraction to a mixed number using a diagram.

Diagrams are not forever. But they are important enough to practice a few times.

2 Convert $\frac{17}{3}$ from an improper fraction to a mixed number using a diagram.

3 Convert $3\frac{1}{5}$ from a mixed number to an improper fraction using a diagram.

4 Convert $2\frac{3}{7}$ from a mixed number to an improper fraction using a diagram.

5 Represent $2\frac{3}{4}$ on a number line.

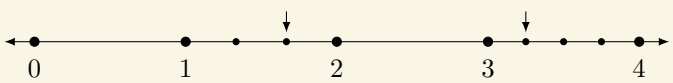
16.2 Fraction Basics - Worksheet 2

1 Convert $\frac{12}{7}$ from an improper fraction to a mixed number using a diagram.

2 Suppose you are given the fraction $\frac{a}{b}$ where a and b are both integers and $b \neq 0$. Describe a calculation that would give you the corresponding mixed number without drawing out a diagram.

Do not divide by zero!

3 Determine the values corresponding to the positions indicated in the diagram below.



4 Completely reduce the fractions $\frac{21}{28}$ and $\frac{18}{48}$.

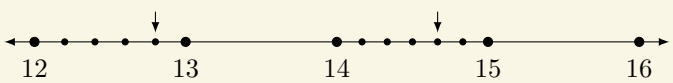
5 Completely reduce the fractions $\frac{8x^2}{6x^4}$ and $\frac{15x^5}{35x^2}$.

16.3 Fraction Basics - Worksheet 3

1 Convert $2\frac{3}{8}$ from a mixed number to an improper fraction using a diagram.

2 Suppose you are given the fraction $a\frac{b}{c}$ where a, b, and c are all integers and $c \neq 0$. Describe a calculation that would give you the corresponding improper fraction without drawing out a diagram.

3 Determine the values corresponding to the positions indicated in the diagram below.



4 Completely reduce the fractions $\frac{25}{40}$ and $\frac{12}{27}$.

5 Completely reduce the fractions $\frac{10x^3y^2}{25xy^5}$ and $\frac{21a^3b^3}{49a^3b^2}$.

16.4 Fraction Basics - Worksheet 4

1 Convert $\frac{23}{5}$ and $\frac{25}{7}$ from improper fractions to mixed numbers without drawing a diagram.

2 Convert $3\frac{2}{9}$ and $4\frac{4}{5}$ from mixed numbers to improper fractions without drawing a diagram.

3 Completely reduce the fractions $\frac{35}{60}$ and $\frac{28m^2n^3}{36mn^8}$.

4 Check the presentation for errors. If you find one, circle it and describe the mistake in words.

$$\frac{4xy}{12x^2y^3} = \frac{4xy}{3xy^2 \cdot 4xy} = \frac{\cancel{4xy}}{3xy^2 \cdot \cancel{4xy}} = 3xy^2$$

This mistake is fairly common.

5 Check the presentation for errors. If you find one, circle it and describe the mistake in words.

$$\frac{20x^2}{5x^5} = \frac{4 \cdot 5x^2}{x^3 \cdot 5x^2} = \frac{4 \cdot \cancel{5x^2}}{x^3 \cdot \cancel{5x^2}} = 4x^3$$

16.5 Fraction Basics - Worksheet 5

1 Completely reduce the fractions $\frac{4x^2}{6x^4}$ and $\frac{4(x+2)^2}{6(x+2)^4}$.

There is no conceptual difference between these two calculations.

2 Completely reduce the fractions $\frac{12xy^4}{18x^4y^2}$ and $\frac{12(x+2)(x-3)^4}{18(x+2)^4(x-3)^2}$.

Do you understand the pattern?

3 Completely reduce the fractions $\frac{9ab^3}{12a^2b}$ and $\frac{9\sin(x)\cos^3(x)}{12\sin^2(x)\cos(x)}$.

Note: $\sin^n(x)$ is a shorthand for $(\sin(x))^n$ and $\cos^n(x)$ is a shorthand for $(\cos(x))^n$.

4 Completely reduce the fraction $\frac{x^2+5x+6}{x^2-3x-10}$.

Hint: Factor, then trust your experience to guide you from there.